RF Sim for KiCad – verification with the TI CC-Antenna-DK2

Simulation vs. TI's measurements (SWRA496A), and the new radiation pattern view

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Summary

Eight PCB antenna boards of TI's antenna kit were simulated with the plugin and compared with TI's measurements: SWR, feed impedance, total efficiency, directivity and, new, the full 3D pattern. The simulation used the automatic mesh and the parts of TI's bill of materials. Nothing was tuned to the measurement.

- **Agrees:** the resonance frequencies (most within 1 % to 4 %), the total efficiency (14 of 19 values within TI's ±3 dB chamber uncertainty), the reactance of the bare antennas #4 and #12, and the 3D pattern in five of ten cases (correlation 0.6–0.9).
- **Does not agree:** the feed resistance is too low (about half at 2.4 GHz, up to a factor 5–7 on the 490 MHz board), so the simulated bands are narrower. TI's directivity is 4 dBi to 6 dBi on every board; the simulation gives 1.8 dBi to 2.4 dBi on most of them.
- **Why:** TI measured the boards with the coax cable and the positioner arm in place. On small boards the cable is part of the radiator: board #1 radiates at 868 MHz only with the cable. TI's own pattern tables show this: at the poles, where the value must not depend on the rotation, they vary by up to 11 dB. TI's directivity is raised further by the scatter of the measurement samples. The simulation describes the bare board.
- **Plugin:** found and fixed during the work: a mesh short of an antenna behind a 0R, a numerical instability with low-loss inductors. New: mesh convergence check, BOM variants, and the radiation pattern in the style of a chamber report (CTIA, as TI) with CST-style cuts.

1 What was compared

Board	Antenna	Board (mm)	Band	Setups
#1	ring antenna, dual-band BOM	26.5 (circle)	868 MHz + 2.4 GHz	as built
#2	compact meander	35 × 45	2.4 GHz	as built
#4	PCB helix, 470–510 MHz BOM	48 × 95	433 MHz to 510 MHz	as built, bare antenna
#6	compact PCB helix	26 × 95	868/915 MHz	as built, bare antenna
#9	dual-band meander, both layers	42 × 95	868 MHz + 2.4 GHz	as built
#10	dual-band helix, 433 MHz BOM	48 × 95	433 MHz	as built
#11	inverted F (TI's recommended design)	30 × 95	2.4 GHz	as built
#12	single-sided compact meander	26 × 95	868/915 MHz	as built, bare antenna

- **Model:** copper and vias from TI's design files, FR-4 1.47 mm with ε_r 4.2 (TI's value), the matching parts as lumped elements with their BOM values (inductor Q 30 at 868 MHz), feed as a 50 Ω port at the connector signal pad, automatic mesh.
- **Bare antenna:** TI also measured the impedance without matching parts. The setups *bare antenna* do the same: series parts 0R, shunt parts not fitted (new BOM variants).
- **TI's data:** SWR plots and impedance markers (network analyser), CTIA over-the-air results (total radiated power at 0 dBm in, so the efficiency includes the mismatch), and the measured 3D pattern tables on a 15° grid.

2 Results

2.1 Resonance and band

Board	Simulation	TI	Deviation
#1 low band	SWR minimum 880 MHz	866 MHz	+1.6 %
#1 high band	SWR minimum 2 370 MHz	2 430 MHz	−2.5 %
#2	SWR 2 band 2442–2661 MHz	2331–2590 MHz	+4.8 / +2.7 %
#4	SWR minimum 487 MHz	about 490 MHz	−0.6 %
#6	SWR 2 band 811–842 MHz	about 843–880 MHz	about −4 %
#9 (4 dips)	850 / 1450 / 1830 / 2670 MHz	860 / 1393 / 1678 / 2520 MHz	−1.2 / +4.1 / +9.1 / +6.0 %
#10	SWR 1.25 at 433 MHz, band 430–436 MHz	SWR 1.27, 9 MHz wide	0 %
#11	SWR 2 band 2460–2638 MHz	2295–2663 MHz	+7.2 / −0.9 %
#12	SWR 2 band 882–902 MHz	about 846–894 MHz	about +2.5 %

The sub-GHz resonances agree within 0 % to 4 %. On the 2.4 GHz meander and inverted-F boards (#2, #9, #11) the simulated resonances are 3 % to 9 % high; the high band of #1 is 2.5 % low. The upper band edges agree better than the lower ones because the simulated bands are narrower. Board #10 is the best case: the match at 433 MHz is the same as TI's.

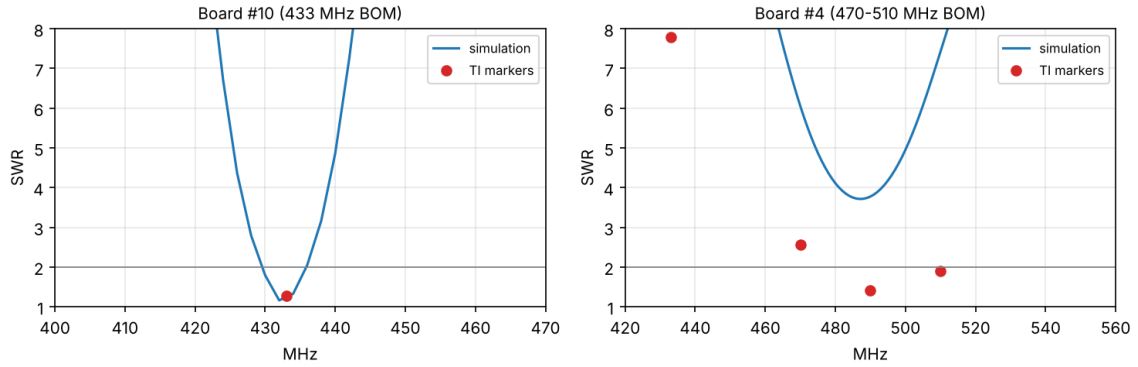


Figure 1: SWR as built, simulation and TI's markers. Left: board #10, the match at 433 MHz is TI's. Right: board #4, the resonance is at TI's frequency but much narrower (section **??**).

2.2 Feed impedance and bandwidth

Board	Frequency	Simulation (Ω)	TI (Ω)
#11 as built	2 440 MHz	$23.2 + j8.0$	$55.2 + j0.5$
#2 as built	2 440 MHz	$26.0 - j9.5$	$41.8 - j1.6$
#9 as built	2 440 MHz	$16.1 + j6.3$	$37.0 - j9.0$
#6 bare antenna	868 MHz	$18.2 + j10.4$	$21.4 - j15.9$
#12 bare antenna	868 MHz	$5.4 - j25.8$	$13.9 - j30.6$
#4 bare antenna	433 MHz	$3.3 - j82.0$	$21.9 - j95.9$
#4 bare antenna	490 MHz	$4.8 - j41.3$	$26.2 - j52.6$

- The reactance of the bare antennas #4 and #12 agrees within 5 Ω to 15 Ω: the electrical length of the antennas is right. On #6 it differs by 26 Ω, as its resonance is 4 % low.
- The resistance is too low on every board except #6. A low resistance means a high Q: the simulated SWR 2 bands are half to two thirds as wide as TI's (#11: 179 vs. 369 MHz).
- The gap grows when the board gets small in wavelengths: about ×2 at 2.4 GHz, ×2.6 on #12 (868 MHz), ×5–7 on #4 (490 MHz, board 0.16 wavelengths long). On #4 the matching network then transforms the wrong resistance, and the simulated SWR minimum stays at 3.8.

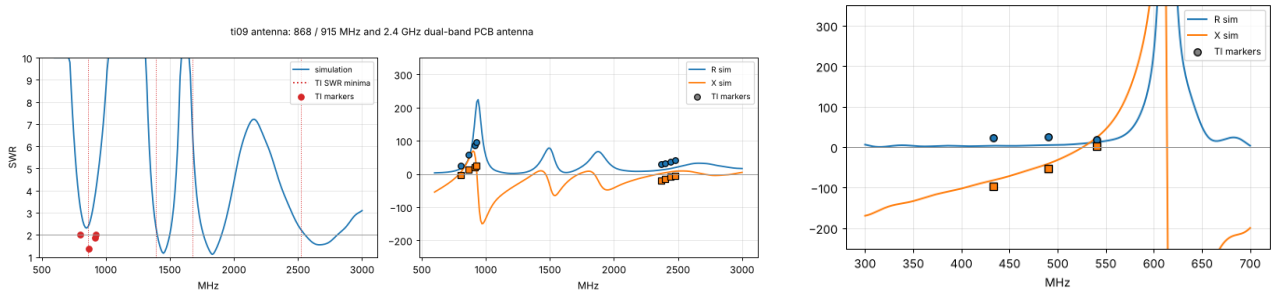


Figure 2: Left: board #9, SWR and impedance from 0.7 to 3 GHz; TI's four SWR minima as dotted lines. Right: board #4, bare antenna: the reactance agrees, the resistance is 5–7 times lower than TI's.

2.3 Efficiency

Board	Frequencies	Total efficiency, simulation	TI
#2	2400 / 2440 / 2480 MHz	67 / 73 / 77 %	64 / 62 / 71 %
#11	2400 / 2440 / 2480 MHz	66 / 73 / 79 %	71 / 69 / 77 %
#9	868 / 915 / 2440 MHz	77 / 60 / 69 %	80 / 43 / 67 %
#6	868 / 915 MHz	49 / 24 %	63 / 29 %
#12	868 / 915 MHz	42 / 52 %	46 / 31 %
#4	470 / 490 / 510 MHz	22 / 35 / 29 %	15 / 37 / 48 %
#10	433 MHz	17 %	26 %
#1	868 / 2440 MHz	0.02 / 1.3 %	52 / 21 %

14 of 19 values lie within TI's ±3 dB. The outliers (red) sit where the simulated band is shifted against TI's (#9 and #12 at 915 MHz, #4 at 510 MHz), and on board #1. Board #1 cannot reach TI's value as a bare board: by the Chu limit ($ka = 0.24$, $Q_{\text{Chu}} = 76$) TI's 40 MHz SWR 2 bandwidth allows at most about 20 % radiation efficiency at 868 MHz.

2.4 Radiation pattern and directivity

TI's report holds the measured pattern of every board as tables (EIRP at 0 dBm into the feed, 15° grid). Integrating them with TI's elevation as θ gives TI's total radiated power within 0.01 dB, so the grid is read correctly. TI's azimuth runs the other way round ($\varphi = -$azimuth): this mapping gives the highest correlation over all boards. A mirror at the board plane fits equally well, as the thin boards radiate nearly the same to both sides. Figures **??–??** show TI's tables and the simulation in the style of TI's report, on one colour scale.

Board	f (MHz)	Length / λ	Pattern correlation	D TI (dBi)	D sim. (dBi)
#9	2440	0.77	+0.90	4.5	4.6
#11	2440	0.77	+0.64	5.9	4.9
#2	2440	0.37	+0.61	4.8	2.4
#1	2440	0.22	+0.58	5.0	3.5
#6	868	0.27	+0.81	4.1	1.9
#9	868	0.27	+0.72	4.5	2.0
#12	868	0.27	+0.22	4.4	1.8
#4	490	0.16	+0.10	4.5	1.9
#10	433	0.14	+0.19	5.2	1.8
#1	868	0.08	−0.03	4.3	5.2

Correlation: of the dB patterns on TI's grid, weighted with $\sin\theta$. D = peak EIRP − TRP.

- **Shape.** On #2, #6, #9, #11 and #1 at 2.44 GHz the shapes agree (0.6–0.9; #9 at 2.44 GHz: 0.90 and the same directivity). On #4, #10 and #12 they do not (0.1–0.2), nor on #1 at 868 MHz. These are the cases where TI's pole values scatter by 5 dB to 11 dB (next point). The simulated bare boards there are short dipoles along the board with a null along $\pm y$; TI's patterns have no such null (figure **??**).
- **The measurement includes more than the board.** A pole ($\theta = 0$ or 180°) is one direction, so its value must not change when the positioner turns the board around the z axis. In TI's tables it changes by 2.5 dB to 3.6 dB on the 2.4 GHz boards and by 8 dB to 11 dB on #4, #10 and #12. So something besides the board changes during the rotation: most likely the cable and the positioner arm, which lies along z. On small boards they radiate.

- **TI's directivity is raised by this scatter.** The peak EIRP is the largest of 312 samples. Adding the scatter seen at TI's poles (standard deviation 0.6 dB to 0.9 dB) to the simulated pattern raises its directivity to TI's value: #11 5.7 vs. 5.9 dBi, #6 4.0 vs. 4.1 dBi, #1 at 2.44 GHz 4.9 vs. 5.0 dBi, #2 4.0 vs. 4.8 dBi. Exception: #9 at 2.44 GHz, which already agrees. On the small boards the scatter is not random (cable), so this simple model does not apply there. The simulated directivity is that of the antenna.

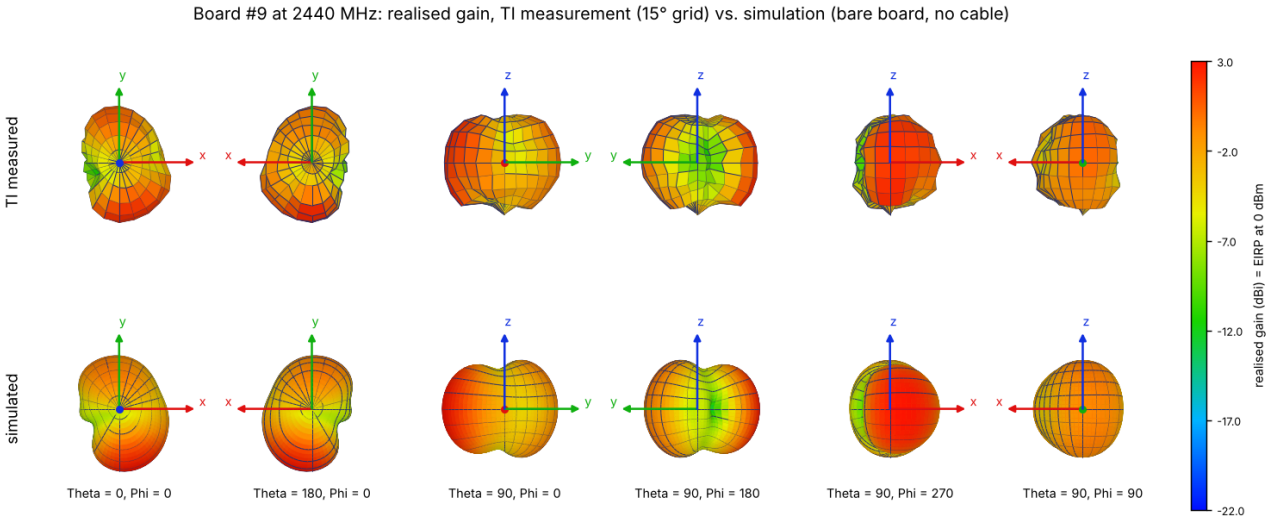


Figure 3: Board #9 at 2 440 MHz, realised gain (= EIRP at 0 dBm in). Top: TI's table, bottom: simulation. The view names are TI's: the direction the pattern is seen from.

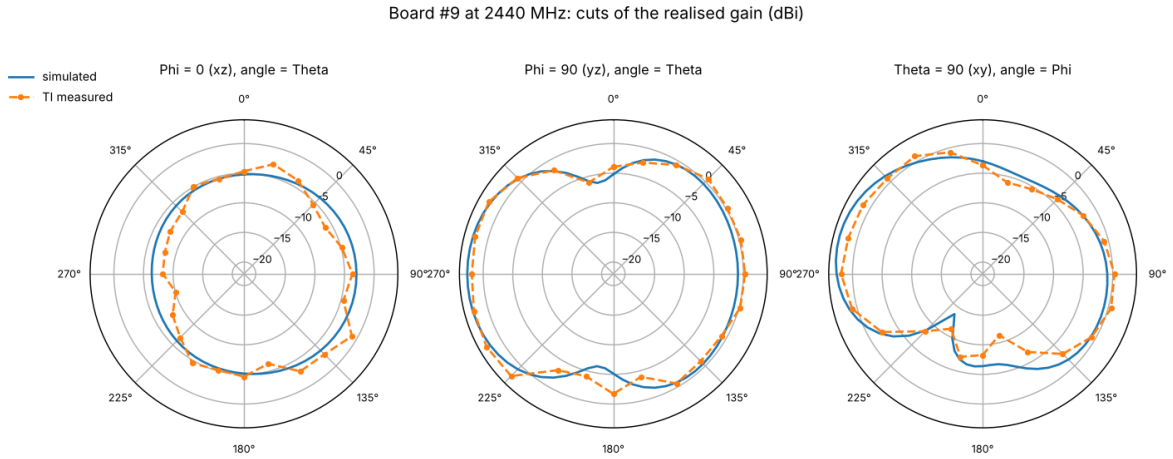


Figure 4: Board #9 at 2 440 MHz: the three principal cuts, TI's samples and the simulation.

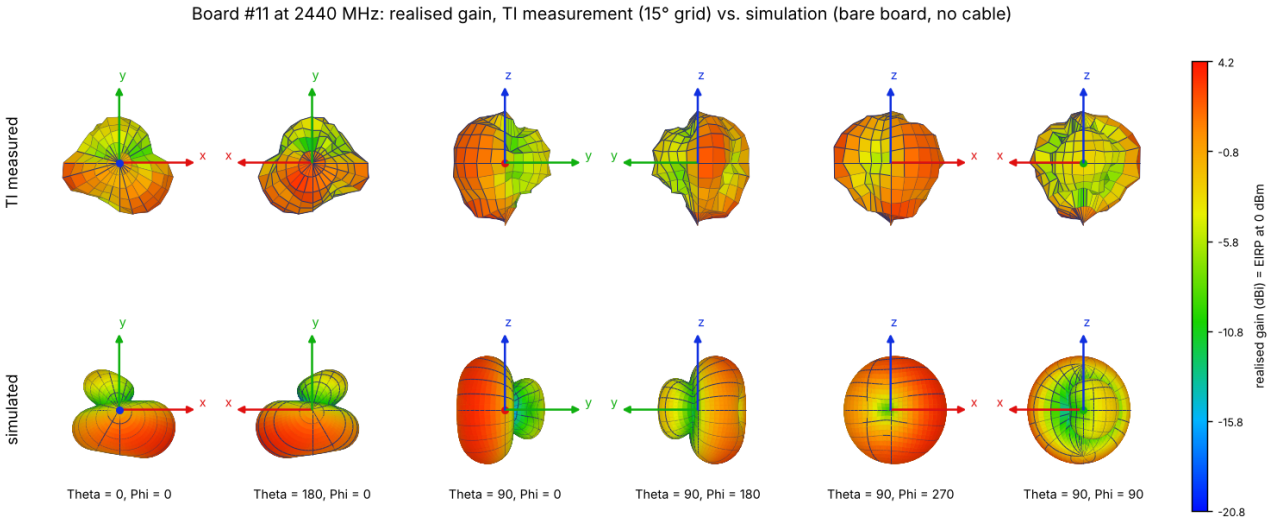


Figure 5: Board #11 (inverted F) at 2 440 MHz. Same lobes; TI's pattern is rougher and has no deep null (correlation 0.64).

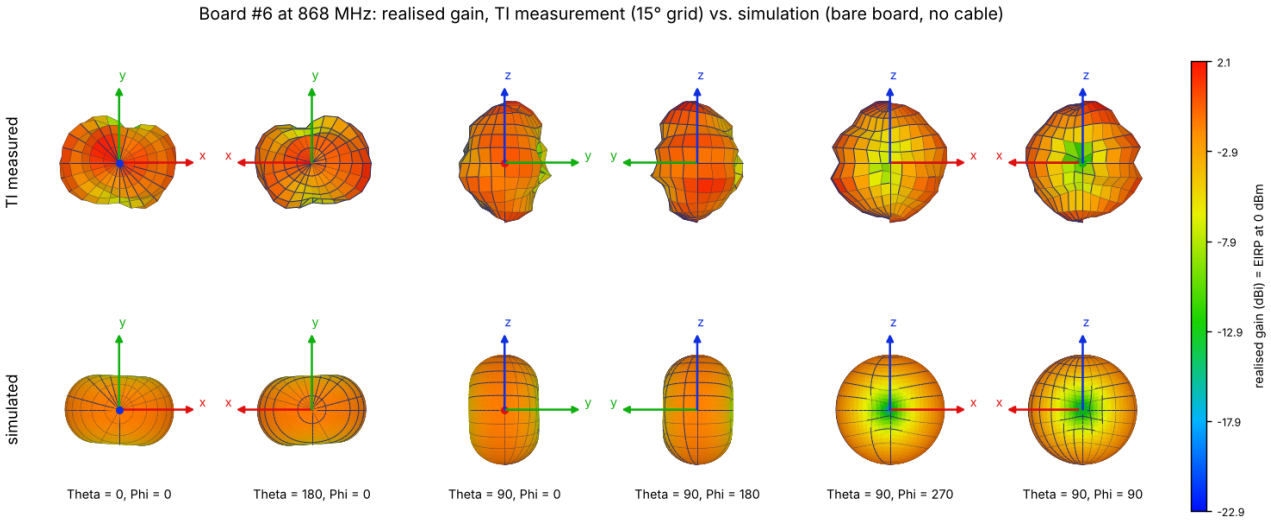


Figure 6: Board #6 (helix) at 868 MHz: same shape (correlation 0.81), TI's surface is rougher.

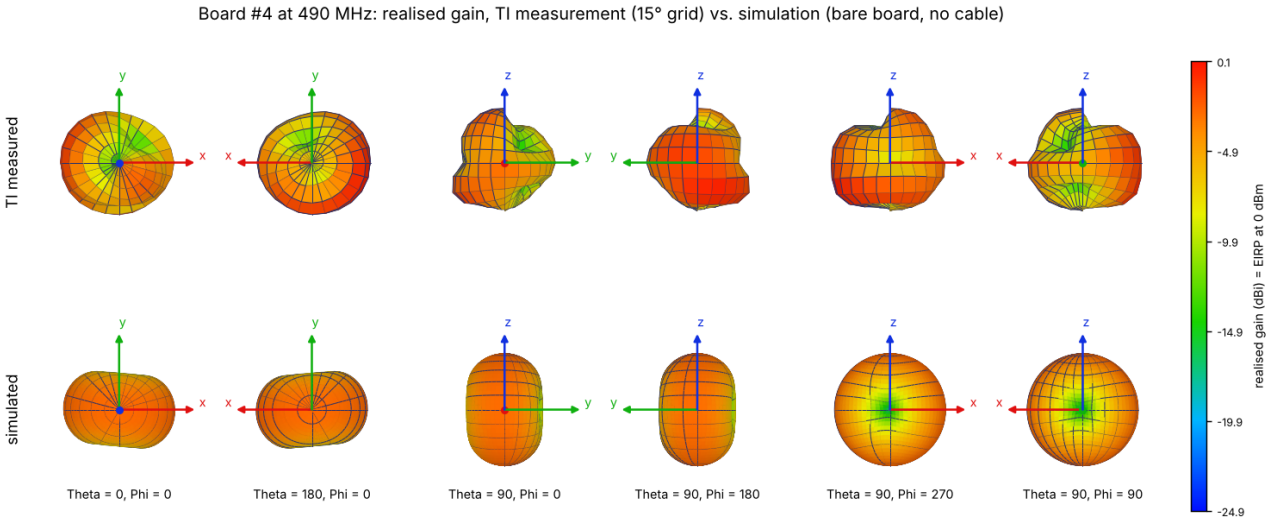


Figure 7: Board #4 at 490 MHz: the simulated bare board is a short dipole along *y*; TI's pattern has no null along *y* (correlation 0.10).

3 Why the simulation and TI differ

3.1 The cable

An experiment added the coax shield to the model: a perfect conductor on the connector ground pads and a 100 mm rod perpendicular to the board.

	No cable	100 mm cable	TI
#1 total efficiency at 868 MHz	0.02 %	36 %	52 %
#1 peak gain at 868 MHz	−33 dBi	−2.5 dBi	1.4 dBi
#1 low-band SWR minimum	880 MHz	830 MHz	866 MHz
#1 total efficiency at 2.44 GHz	1.3 %	1.6 %	21 %
#11 SWR 2 band	2458–2637 MHz	2430–2697 MHz	2295–2663 MHz
#11 Z at 2.44 GHz	$23 + j8\,\Omega$	$26 - j5\,\Omega$	$55 + j0\,\Omega$

- On board #1 the cable is the radiator at 868 MHz; on #11 it widens the band by half.
- The straight rod does not reproduce TI's patterns: the real cable route is not known.
- Measured values of small boards depend on the cable. For a product without a cable the simulation of the bare board is the cleaner reference.

3.2 Other causes

- **Frequency at 2.4 GHz (3 % to 9 % high).** Not modelled: solder mask over the antenna (typically −1 % to −2 %) and an ε_r above 4.2 at 2.4 GHz. ε_r 4.5 moves #11 by −30 MHz (−1.2 %); it does not close the gap alone.
- **Inductor loss.** The default Q 30 at 868 MHz falls to about 11 at 2.44 GHz. TI's Murata LQW15AN parts have Q 40–70 there. With Q 40 and the cable, #1 reaches 43 % at 868 MHz (TI 52 %) and 4 % at 2.44 GHz (TI 21 %). The high band of #1 stays unexplained; the self-resonance of the parts (4 GHz to 6 GHz) is not modelled.
- **Checks that ruled things out.** The board copies match TI's Gerbers. Mesh: Fast vs. Automatic moves the #11 resonance by 0.4 %. Feed port: a vertical port adds about $+j11\,\Omega$; the side port at the connector is the right model for a coplanar feed.

4 Plugin changes made during this work

4.1 Fixes

Problem	Now
Mesh short of an antenna fed through a 0R (found on #9)	The mesh check also guards the antenna copper behind 0R parts and lumped parts. Before, the coarse board mesh joined the antenna stub to the bottom ground pour ($Z \approx j12\,\Omega$).
Clearance lines shrank the cells (found on #9)	Lines added against mesh shorts no longer set the merge distance: no more 0.025 mm cells and halved timestep.
Missed clearance line (found on #12)	A needed line next to an existing mesh line is now added in a second pass ($Z \approx j10\,\Omega$ before). The meshes of the other boards are unchanged.
Instability with low-loss inductors	The stability capacitance of a lumped inductor is now limited by the part's reactance. A run with Q 40 at 2.44 GHz diverged before.

4.2 New

- **Mesh convergence check** (Simulate → *Mesh convergence check*): the setup on Fast, Automatic and optionally Accurate; the key results side by side with tolerances, the extrapolation to zero cell size, and all S11 curves in one chart. Example #11, Fast vs. Automatic: SWR minimum −0.21 %, resonance −0.38 %, radiation efficiency +0.1 points, directivity +0.26 dB, R +3.2 Ω. Verdict *not converged* for the depth of the null and R: near a perfect match the depth is very sensitive.
- **BOM variants:** `"lumped_values": {"Z14": "0R", "Z15": "DNP"}` in an antenna setup replaces part values. The unit decides the kind. A 0R on any footprint is a short.
- **Radiation pattern in the style of a chamber report** (section **??**).
- **Polarisation:** the far field is stored as E_θ and E_φ parts. Older results get them from the kept far-field file.
- **Pattern sheets:** every run writes `pattern_<f>MHz.png` per far-field frequency.

5 The new radiation pattern view

The antenna report (the 360° view) now draws the pattern like TI's CTIA report and CST:

- **Frame:** x right, y up on the board, z out of the top side; θ from $+z$, φ from $+x$ towards $+y$. This is TI's frame and CST's default.
- **Quantity:** directivity, or realised gain. With 0 dBm into the feed the realised gain in dBi is the EIRP in dBm of a CTIA report. Polarisation: total, E_θ or E_φ.
- **Style** *chamber report*: blue-green-yellow-red scale, 15° wire grid, $x/y/z$ arrows, colour bar with five values, range *min to max* as in TI's report.
- **Views:** free 3D, the six views of a chamber report on one page, or one of them; drawn orthographic like the report. *Over the board* shows the pattern around the antenna copper.
- **Cuts** as in CST: $\varphi = 0$ (xz), $\varphi = 90$ (yz), $\theta = 90$ (xy) with total, E_θ and E_φ, main lobe magnitude and direction, 3 dB angular width and side lobe level.

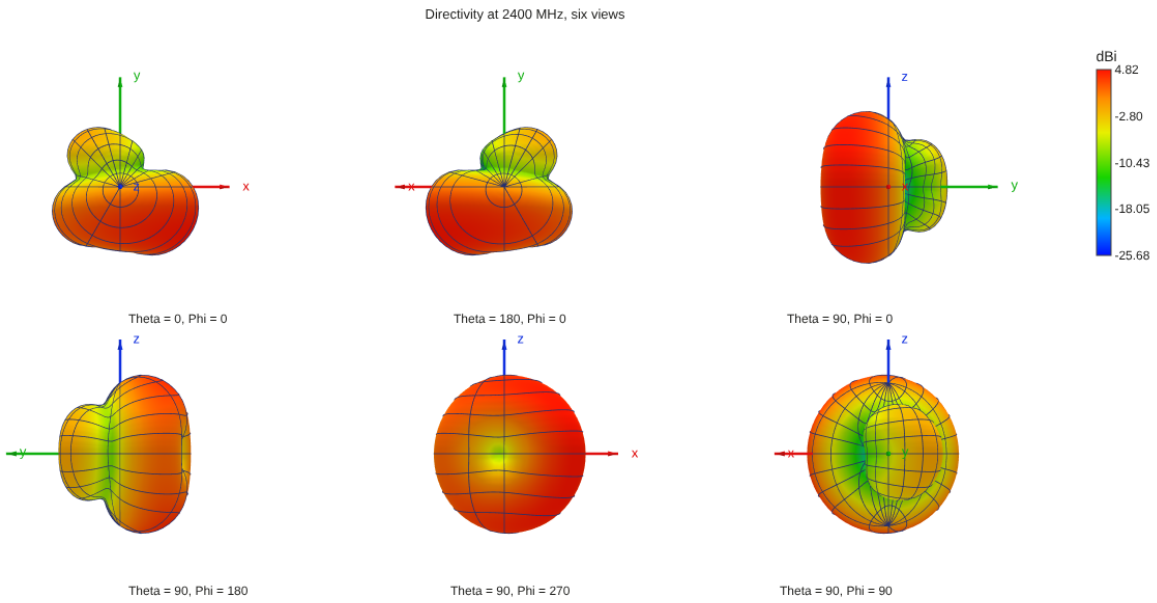


Figure 8: Antenna report, view *six views (report)*: board #11 at 2 400 MHz.

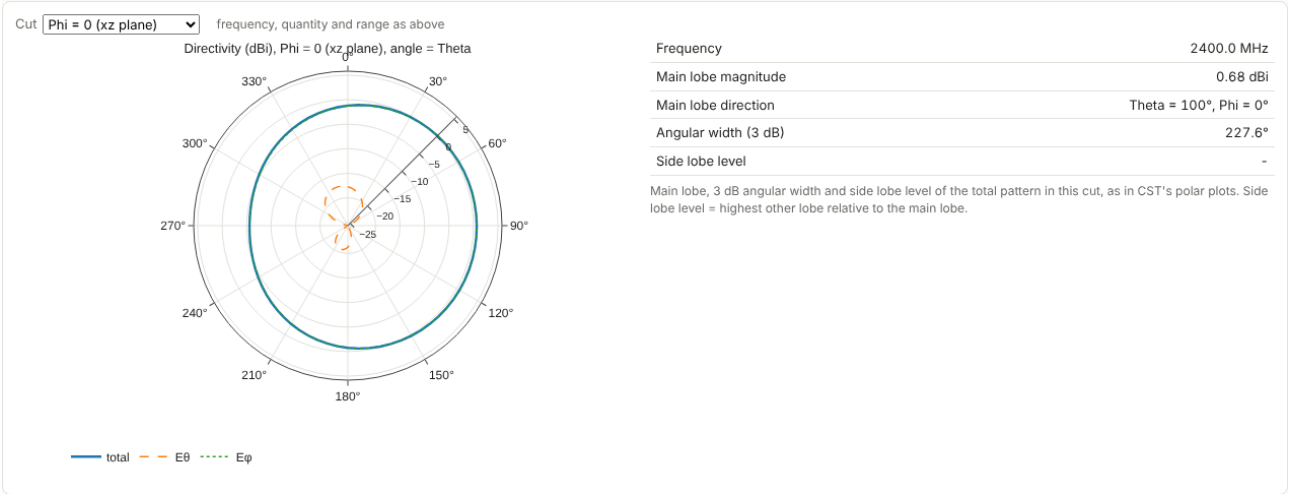


Figure 9: Antenna report, cut card: polar cut with E_θ / E_φ and the main lobe values.

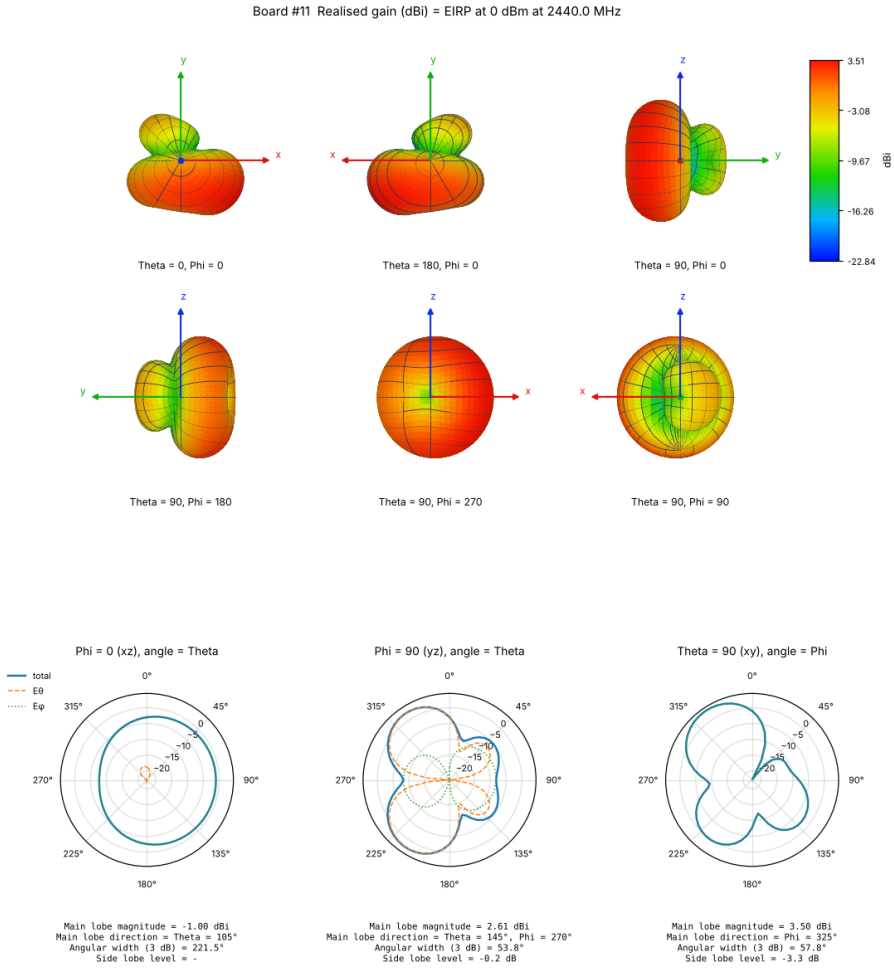


Figure 10: Pattern sheet written by every run (`pattern_2440MHz.png`), board #11.

6 Recommendations

- **Trust** the simulated resonance, reactance, efficiency and pattern of the bare board. Compare small boards with a measurement only when the cable is in the model or choked (ferrites, a cable along the null of the antenna).
- **Measuring your own boards:** keep the cable away from the antenna and lead it along the ground plane. Check the pole values of a chamber pattern: a large spread there means that the fixture radiates.
- **Plugin, next:** a frequency-dependent inductor model (vendor Q and self-resonance) in the antenna FDTD, and an optional cable model (shield as a conductor with a chosen route).
- **Release 0.25.0** with the fixes and features of this report.